

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement #14 - Healthcare & Telemedicine - Corporate Mental Wellness & Counseling Platform

Deliverable 1: Requirements Table

Scenario: An anonymous counselling portal for a company. Employees can check their stress levels, book confidential sessions with a licensed therapist and follow their own wellness indicators, and at no point does the employer learn who used it.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall issue an anonymised pseudo-ID to every employee account and use only that pseudo-ID in the booking data shown to a therapist, so that no corporate email address or employee ID is disclosed.	High	Pass: after an employee books a session, the therapist dashboard and the stored booking record show only the pseudo-ID in the format WEL-XXXXXX. Fail: any booking metadata field contains the employee's email, employee ID or full name.	The whole service depends on this. An employee who thinks HR can see their name simply will not book.
FR-002	Functional	The system shall let an employee complete a stress self-assessment questionnaire and shall return a numeric score together with its severity band (low, moderate, high or severe) once the final question is answered.	High	Pass: submitting a fully answered questionnaire returns a score between 0 and 27 and the band matching the published score ranges, within 2 seconds. Fail: no result is returned, the band contradicts the score, or a partly answered form is scored.	Gives the employee and the therapist the same starting point. The score is also what FR-005 checks against.
FR-003	Functional	The system shall allow an employee to book a counselling session only in a slot the therapist has published as free, and shall mark that slot unavailable in the same transaction.	High	Pass: when two employees confirm the same slot at the same moment, one booking is confirmed and the other receives a 'slot no longer available' message. Fail: both bookings are confirmed, or a slot outside the therapist's published availability is accepted.	Stops double booking. A therapist who finds two people waiting for the same slot has to turn one of them away.
FR-004	Functional	The system shall show an employee a trend of their own wellness indicators (stress score, sleep hours and sessions attended) across their six most recent assessments.	Medium	Pass: an account with six or more assessments plots all six in date order; an account with fewer plots only the assessments that exist. Fail: the chart shows another account's data, or fills dates that have no assessment with estimated values.	A trend is easier to read than a list of old scores, and it gives the employee a reason to keep taking the assessment.

FR-005	Functional	The system shall, whenever an assessment falls in the severe band, display regional crisis helpline details on the results screen and offer the earliest available priority appointment slot.	High	Pass: a submission scoring in the severe band shows the helpline panel and lists at least one priority slot starting within the next 24 hours. Fail: a severe score is presented as an ordinary result, with no helpline and no priority slot.	A severe score needs attention the same day, not after a two-day wait in the normal booking queue.
NFR-001	Nonfunctional (Performance & Security)	The appointment scheduler shall convert every slot to the viewer's local timezone correctly, including across daylight saving transitions, and shall return a slot list in under 2 seconds at peak load.	High	Pass: a benchmark of 500 concurrent users across IST, GMT and PST keeps 95th percentile slot-list latency below 2 seconds, and a booking made on a daylight saving changeover date shows the same absolute time to both parties. Fail: the two views disagree on the time, or 95th percentile latency exceeds 2 seconds.	Employees and therapists sit in different countries and book across timezones daily. An hour's error means somebody misses the session.
NFR-002	Nonfunctional (Security & Privacy)	The system shall store session notes and assessment answers encrypted with AES-256 at rest and TLS 1.3 in transit, and shall release an aggregate wellness report to a Wellness Program Administrator only where the reported group contains at least ten employees.	High	Pass: a database dump contains no readable note text, and a report requested for a department of nine employees is refused with a 'group too small' message. Fail: plaintext notes appear in storage or logs, or a report is released for a group smaller than ten.	Session notes are clinical records. Reports on very small teams would also let HR work out who has been using the service.

Traceability: requirement to use case

Use Case	Title	Traces to
UC-01	Create Anonymous Wellness Account	FR-001
UC-02	Take Stress Self-Assessment	FR-002
UC-03	Book Counseling Session	FR-003
UC-04	View Wellness Trend Dashboard	FR-004
UC-05	Manage Availability Calendar	FR-003
UC-06	Record Encrypted Session Notes	NFR-002
UC-07	View Aggregate Wellness Report	NFR-002
UC-08	Generate Pseudo-ID	FR-001
UC-09	Escalate to Crisis Support	FR-005
UC-10	Verify Slot Availability	FR-003
UC-11	Send Confirmation Notification	FR-003, NFR-001

UC-12	Reschedule or Cancel Session	FR-003
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